which they are immersed during the time that elapses between the start of the shot and the end of the process. Space and time are the gravitational field, Einstein's 'mollusc': we are also including the gravitational field, that is to say, a piece of the mollusc, in the process. Everything is immersed in Einstein's great mollusc: here, imagine that you are slicing a small, finite portion of it, like a piece of sushi, which encompasses the collision and what surrounds it.

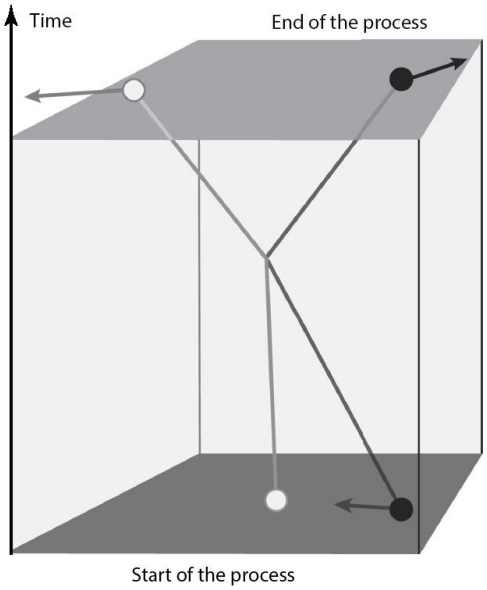


Figure 7.1 A region of space in which a black ball hits a stationary white ball, propels it and rebounds. The box is the region of spacetime. Within it are drawn the trajectories of the balls.

What we obtain from this is a spacetime box (as in figure 7.1): a finite portion of spacetime a few cubic metres in dimension by a few seconds of time. *This* process does not occur 'in' time. The box is not *in* spacetime, it *includes* spacetime. It isn't a process *in* time, in the same way in which grains of space are not *in* space. The passage of time is only the measure of the process itself, just as quanta of gravity are not *in* space, as they themselves constitute space.

The key to understanding how quantum gravity works lies in considering not solely the physical process given by the two balls but rather the entire process defined by the whole box with all that it entails, including the gravitational field.

Now let us return to Heisenberg's original insight: quantum mechanics does not tell us what happens during the course of a process, but the probability which ties together the different initial and final states of the process. In our case, the initial and final states are given by all that happens at the *border* of the spacetime box.